

March 24, 2025

Dear NMR in Biomedicine editorial board,

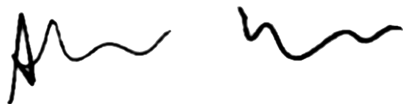
We wish to submit our original manuscript entitled "Assessing Regional Cerebral Oxygen Consumption (CMRO₂) in Preterm Neonates: A Quantitative MRI Cohort Study with Exploratory Analysis of Respiratory Support" for consideration by NMR in Biomedicine. Our study presents a novel multi-modal MRI approach for non-invasively measuring cerebral oxygen metabolism in preterm infants. Given the journal's focus on advancing biomedical applications of NMR, we believe our findings will be of particular interest to your readership.

This work introduces an MRI-based method to quantify cerebral blood flow (CBF) and cerebral metabolic rate of oxygen (CMRO₂) in very preterm infants, a critical step in understanding brain development and vulnerability to injury. Our exploratory analysis suggests that different modes of respiratory support influence cerebral oxygenation, with non-invasive ventilation associated with increased oxygen delivery and utilization. These insights could enhance clinical strategies for neonatal care and long-term neurodevelopmental outcomes.

We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the manuscript and agree with its submission to NMR in Biomedicine. We have no conflicts of interest to disclose.

Thank you for your consideration of our manuscript.

Sincerely,



Dr. Alexander Mark Weber